

Positive Square Energy of Every Bicyclic Cactus

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Abstract

For every connected bicyclic cactus G on n vertices we prove $s^+(G) \geq n$. The statement permits the two cyclic blocks to share a cut vertex or to be joined by a path of arbitrary length, and permits arbitrary trees at every vertex. An exact block-additive doubly nonnegative certificate reduces the problem to cycle-length pairs $\{5, 5\}$ and $\{3, 4k+1\}$. A product-subpartition phase comparison settles the latter. For two pentagons, a coefficient certificate handles the shared-vertex bouquet, while an actual-lobe continuant factorization handles every nontrivial connector. The residual bounds are strict; the omnibus conclusion is stated weakly.

1 Preliminaries

All graphs are finite, simple, and undirected. For the adjacency eigenvalues $\lambda_1, \dots, \lambda_n$ of G , put

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \quad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2, \quad D(G) = s^+(G) - s^-(G).$$

The elementary energy identity is

$$s^+(G) + s^-(G) = \operatorname{tr} A(G)^2 = 2|E(G)|. \quad (1)$$

A connected bicyclic graph has $|E(G)| = n + 1$, so

$$s^+(G) = n + 1 + \frac{1}{2}D(G). \quad (2)$$

Liu–Tang–Zhang proved the positive and negative square-energy conjecture using a doubly non-negative matrix inequality [2]. Akbari–Kumar–Mohar–Pragada–Zhang (AKMPZ) conjecture that (2) is at least n for every connected graph with at least $n + 1$ edges [1]. We prove that assertion here only for bicyclic cacti, not for all bicyclic graphs.

For positive vertex activities $a = (a_v)$ define the signless matching partition

$$Z_F(a) = \sum_{M \text{ matching of } F} \prod_{v \text{ unmatched by } M} a_v, \quad Z_F(t) = Z_F((t)_v). \quad (3)$$

Set

$$\Psi_G(t) = i^{-n} \det(itI - A(G)) = \prod_{j=1}^n (t + i\lambda_j), \quad t > 0. \quad (4)$$

Grouping Sachs' expansion by its collection $\mathcal{C}$ of pairwise vertex-disjoint cycle components gives the normalized Sachs formula

$$\Psi_G(t) = \sum_{\mathcal{C}} \prod_{C \in \mathcal{C}} (-2i^{-|C|}) Z_{G-V(C)}(t). \quad (5)$$

Let

$$\Theta_G(t) = \sum_j \arctan(\lambda_j/t)$$

be the continuous argument of $\Psi_G(t)$ tending to zero at infinity. The signed Coulson identity is

$$D(G) = -\frac{4}{\pi} \int_0^\infty t \Theta_G(t) dt. \quad (6)$$

Indeed, $\lambda|\lambda| = (2/\pi) \int_0^\infty \lambda^3/(\lambda^2 + t^2) dt$; summing, using $\sum_j \lambda_j = 0$, and integrating by parts proves (6). The same cancellation gives $\Theta_G(t) = O(t^{-3})$, so the integral converges.

We repeatedly remove trees exactly. If a branch is oriented toward its attachment and $T_{u \rightarrow p}$ is the subtree below u , then

$$Q_{u \rightarrow p} = \frac{Z_{T_{u \rightarrow p}}(t)}{Z_{T_{u \rightarrow p} - u}(t)} = t + \sum_{w \text{ child of } u} \frac{1}{Q_{w \rightarrow u}} \geq t. \quad (7)$$

Thus a retained spine vertex v receives activity

$$a_v = t + \sum_{u \text{ attached at } v} Q_{u \rightarrow v}^{-1} = t + y_v, \quad y_v \geq 0, \quad (8)$$

and all Sachs terms have the same positive factor $K(t) = \prod Z_{T_{u \rightarrow v}}(t)$. This remains true when a Sachs cycle deletes v : its detached branches contribute their full Z_T factors. Equations (7)–(8) follow by splitting a matching according as its root is unmatched or matched to one child.

2 The sharp cactus DNN reduction

For a connected edge-containing graph define

$$\kappa(G) = \sup_{\substack{M \succeq 0, M \geq 0 \\ \mathbf{1}^T M \mathbf{1} > 0}} \frac{4(\sum_{uv \in E(G)} \sqrt{M_{uv}})^2}{\mathbf{1}^T M \mathbf{1}}. \quad (9)$$

Theorem 2.1 (Sharp cactus DNN lemma). *If a cactus G has b bridge blocks and cyclic blocks of lengths $\ell_1, \dots, \ell_r$, then*

$$\kappa(G) = b + \sum_{j=1}^r \kappa(C_{\ell_j}), \quad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even}, \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd}. \end{cases} \quad (10)$$

Moreover $s^-(G) \leq \kappa(G)$.

Proof. We include the sharp ingredients; full bookkeeping appears in the companion manuscript [3]. Fold every nonedge entry M_{uv} into the two diagonal entries by adding $M_{uv}(e_u - e_v)(e_u - e_v)^T$. This preserves the denominator and edge entries and leaves an edge-supported DNN matrix.

On C_ℓ , rotational averaging and concavity of $\sqrt{\cdot}$ do not decrease the objective. If t is the common edge entry and r the row sum, the objective is $4\ell t/r$. A Fourier eigenvalue gives $r \geq 4t$ for even ℓ , and $r \geq 2(1 + \cos(\pi/\ell))t$ for odd ℓ . Equality is attained by $x(2cI + A(C_\ell))$, where $c = 1$ in the even case and $c = \cos(\pi/\ell)$ in the odd case. This proves the cycle constants; $\kappa(K_2) = 1$ follows from $a + d \geq 2c$ for a DNN $\begin{pmatrix} a & c \\ c & d \end{pmatrix}$.

If $G = G_1 \vee_z G_2$, let g_v be Gram vectors after folding, and put

$$U = \text{span}\{g_v : v \in V(G_1) - z\}, \quad W = \text{span}\{g_v : v \in V(G_2) - z\}.$$

All cross-side pairs are nonedges, so $U \perp W$. Decompose the cut vector orthogonally as $g_z = u + w + h$, with $u \in U$, $w \in W$, and $h \perp U + W$. On G_1 use $u + h$ as its cut vector and on G_2 use w ; all other vectors remain unchanged. This preserves every edge inner product and gives DNN Gram matrices. If

$$a = \sum_{v \in V(G_1) - z} g_v, \quad b = \sum_{v \in V(G_2) - z} g_v,$$

then $a + u \in U$, $b + w \in W$, and these vectors are orthogonal to each other and to h . Consequently

$$\begin{aligned} T(G) &= \|a + b + u + w + h\|^2 \\ &= \|a + u + h\|^2 + \|b + w\|^2 = T_1 + T_2. \end{aligned}$$

Also $S = S_1 + S_2$. Hence

$$2(S_1 + S_2) \leq \sqrt{\kappa_1 T_1} + \sqrt{\kappa_2 T_2} \leq \sqrt{(\kappa_1 + \kappa_2)(T_1 + T_2)}.$$

Conversely glue near-optimal Gram systems orthogonally at z and scale them so that $T_1/\kappa_1 = T_2/\kappa_2$. Thus $\kappa(G_1 \vee_z G_2) = \kappa_1 + \kappa_2$, proving block additivity.

Finally write $A = P - B$ for the positive and negative spectral parts. The Schur product theorem makes $M = B \circ B$ DNN, while

$$s^-/2 = - \sum_{uv \in E} B_{uv} \leq \sum_{uv \in E} |B_{uv}|, \quad \mathbf{1}^T M \mathbf{1} = \text{tr } B^2 = s^-.$$

Substitution in (9) gives $(s^-)^2 \leq \kappa(G)s^-$ and hence $s^- \leq \kappa(G)$. □

Define

$$\epsilon_\ell = \begin{cases} 0, & \ell \text{ even}, \\ \ell \tan^2(\pi/(2\ell)), & \ell \text{ odd}. \end{cases} \quad (11)$$

Thus $\kappa(C_\ell) = \ell + \epsilon_\ell$. Through odd integers $\ell \geq 3$, ϵ_ℓ is strictly decreasing: with $u = \pi/(2x)$ this is equivalent to monotonicity of $\tan^2 u/u$, whose derivative is positive because $2u > \sin u \cos u$. Moreover

$$\epsilon_3 = 1, \quad \epsilon_5 = 5 - 2\sqrt{5}, \quad \epsilon_5 + \epsilon_7 < 1. \quad (12)$$

For the last strict inequality, $\cos(\pi/7) > 7/8$ gives $\epsilon_7 < 7/15 < 2\sqrt{5} - 4 = 1 - \epsilon_5$.

If the two cycle lengths are p, q , then the number of bridge blocks satisfies $b + p + q = n + 1$, and Theorem 2.1 gives

$$s^+(G) = 2n + 2 - s^-(G) \geq n + 1 - \epsilon_p - \epsilon_q. \quad (13)$$

Consequently DNN settles every pair containing an even length and every pair of odd lengths at least five except $\{5, 5\}$. For two cycle lengths both 3 (mod 4), formula (5) has strictly negative imaginary part: each singleton cycle has multiplier $-2i$, whereas the empty and possible double-cycle terms are real. Hence $\Theta_G(t) < 0$ and

$$D(G) > 0, \quad s^+(G) > n + 1. \quad (14)$$

After these arguments exactly $\{5, 5\}$ and $\{3, 4k + 1\}$ remain.

3 A triangle and a 1 (mod 4) cycle

Theorem 3.1. *Suppose the cyclic blocks of G have lengths 3 and $q = 4k + 1 \geq 5$. With arbitrary incidence and arbitrary attached trees,*

$$s^+(G) > n + 2 - \sec(\pi/q) > n.$$

Proof. Let H be the union of the cycles and, when they are disjoint, their unique joining path. Eliminate every off-spine tree by (7)–(8). With all partitions below using the effective activities, put

$$A = Z_{H-V(C_3)}(a), \quad B = Z_{H-V(C_q)}(a),$$

and in the disjoint case put $D_0 = Z_{H-(V(C_3) \cup V(C_q))}(a)$. The cycle multipliers are $-2i$ and $2i$. Thus

$$\Psi_G/K = R + 2i(B - A), \quad R = \begin{cases} Z_H(a), & \text{shared vertex,} \\ Z_H(a) + 4D_0, & \text{disjoint.} \end{cases} \quad (15)$$

In particular $R > 0$ and $\Theta_G = \arctan(2(B - A)/R)$.

Let $Z_q(t) = Z_{C_q}(t)$. Partition matchings counted by $Z_H(a)$ according as they use a boundary edge between $S = V(C_q)$ and its complement. The class using none factors, also in the shared case, and therefore

$$Z_H(a) = Z_{C_q}(a|_S)B + E, \quad E \geq 0. \quad (16)$$

Since every activity is $t + y_v$ and matching partitions have nonnegative coefficients, write $Z_{C_q}(a|_S) = Z_q(t) + L$ with $L \geq 0$. Equations (15)–(16) give the full product-subpartition comparison

$$R - Z_q(t)(B - A) = E + LB + Z_q(t)A (+4D_0) > 0, \quad (17)$$

where the parenthesized term occurs only when the cycles are disjoint. If $B - A \leq 0$ the desired phase comparison is immediate; otherwise divide (17) by $RZ_q(t)$. In both cases

$$\Theta_G(t) < \arctan \frac{2}{Z_q(t)} = \Theta_{C_q}(t). \quad (18)$$

The eigenvalues $2 \cos(2\pi j/q)$ and the finite cosine sum give, for $q = 4k + 1$,

$$s^+(C_q) = q + 1 - \sec(\pi/q), \quad D(C_q) = -2(\sec(\pi/q) - 1).$$

Integrating the strict pointwise inequality (18) in (6) yields $D(G) > D(C_q)$. Equation (2) now gives $s^+(G) > n + 2 - \sec(\pi/q) > n$, since $q \geq 5$ and $\sec(\pi/q) < 2$. $\square$

This proof is the product-subpartition argument of the companion manuscript *The Triangle- C_{4k+1} Residual Family for Bicyclic Cacti* [4]; no restriction is placed on the connector or trees.

4 Two pentagons

4.1 The shared-vertex bouquet

Theorem 4.1. *If the two C_5 blocks share one vertex and arbitrary trees are attached, then*

$$s^+(G) \geq n + 1 - \frac{4}{3\sqrt{13}} > n.$$

Proof. We give the exact finite certificate behind the proof in [5]. After tree elimination, label the common activity a_0 and the four activities on lobe j by $x_{j,1}, \dots, x_{j,4}$. For one lobe define

$$\begin{aligned} A(x) &= x_1 x_2 x_3 x_4 + x_3 x_4 + x_1 x_4 + x_1 x_2 + 1, \\ B(x) &= x_2 x_3 x_4 + x_2 + x_4 + x_1 x_2 x_3 + x_1 + x_3. \end{aligned}$$

Here A is the partition after deleting the common vertex, while B sums the two states matching that vertex into the lobe. Writing A_j, B_j for the two lobes,

$$\Psi_G/K = R + 2i(A_1 + A_2), \quad R = a_0 A_1 A_2 + B_1 A_2 + A_1 B_2. \quad (19)$$

The exact coefficient certificate is

$$2R \geq t(t^4 + 7t^2 + 9)(A_1 + A_2) \quad \text{whenever all nine activities are at least } t. \quad (20)$$

For reproducibility, substitute $a_0 = t + y_0$ and $x_{j,k} = t + y_{j,k}$ into the difference between the two sides. Its expansion over $\mathbb{Z}$ has exactly 1290 nonzero monomials, every coefficient lies in $\{1, \dots, 22\}$, and every monomial contains a y variable. The canonical coefficient stream has SHA-256 digest

4c436cac772395d2a8edfdd81408ffe426759d3e94d66df2e4ab0235a3343110.

Appendix A records the exact script and command. This coefficientwise statement proves (20) on the entire nonnegative orthant, rather than by sampling.

From (19)–(20),

$$0 < \Theta_G(t) \leq \arctan \frac{4}{t(t^4 + 7t^2 + 9)}.$$

Using $\arctan u \leq u$ and the reciprocal-quartic integral gives

$$\int_0^\infty t \Theta_G(t) dt \leq \int_0^\infty \frac{4}{t^4 + 7t^2 + 9} dt = \frac{2\pi}{3\sqrt{13}}.$$

Thus $D(G) \geq -8/(3\sqrt{13})$ by (6), and (2) proves the theorem. $\square$

4.2 Every nontrivial connector

Theorem 4.2. *If the two C_5 blocks are vertex-disjoint, joined by their unique path of arbitrary positive length, and arbitrary trees are attached anywhere, then*

$$s^+(G) > n + 5 - 2\sqrt{5} > n.$$

Proof. Let the joining path have endpoints u_i on the pentagons and $m \geq 0$ internal vertices. Let X_i be the *actual lobe*: the i th pentagon together with every tree attached at its cycle vertices, rooted at u_i . Eliminate only branches at internal path vertices, obtaining positive activities $a_1, \dots, a_m$ and a positive factor B_0 . Put

$$z_i = \Psi_{X_i}(t), \quad w_i = \Psi_{X_i - u_i}(t), \quad r_i = w_i/z_i.$$

The graph $X_i - u_i$ is a forest, and Sachs gives

$$z_i = Z_{X_i}(t) + 2iZ_{X_i - V(C_5)}(t).$$

Hence z_i is in the open first quadrant and $r_i = x_i - iy_i$ with $x_i, y_i > 0$.

For the continuant

$$K() = 1, \quad K(b_1) = b_1, \quad K(b_1, \dots, b_k) = b_k K(b_1, \dots, b_{k-1}) + K(b_1, \dots, b_{k-2}),$$

the last-edge split of path matchings gives the exact actual-lobe factorization

$$\Psi_G = B_0 z_1 z_2 F, \quad F = \begin{cases} 1 + r_1 r_2, & m = 0, \\ a_1 + r_1 + r_2, & m = 1, \\ K(a_1 + r_1, a_2, \dots, a_{m-1}, a_m + r_2), & m \geq 2. \end{cases} \quad (21)$$

To see both exactness and the phase sign, expand

$$F = A + Br_1 + Cr_2 + Dr_1 r_2. \quad (22)$$

The four coefficients are the path matching partitions obtained by prescribing neither, only the left, only the right, or both boundary edges. For $m = 0$, $(A, B, C, D) = (1, 0, 0, 1)$; for $m = 1$ it is $(a_1, 1, 1, 0)$; for $m \geq 2$ all four are positive. Therefore

$$\text{Im } F = -By_1 - Cy_2 - D(x_1 y_2 + x_2 y_1) < 0. \quad (23)$$

Let $\theta_i = \text{Arg } z_i \in (0, \pi/2)$. Formula (5) for G has positive imaginary part (the two singleton pentagon terms), so $0 < \Theta_G < \pi$. Equations (21)–(23), with the continuous branch, imply

$$0 < \Theta_G = \theta_1 + \theta_2 + \text{Arg } F < \theta_1 + \theta_2. \quad (24)$$

Indeed $\text{Arg } F \in (-\pi, 0)$, and the alternative branch contradicts $0 < \Theta_G < \pi$. Coulson integration now yields

$$D(G) > D(X_1) + D(X_2). \quad (25)$$

It remains to bound an actual lobe X . Eliminate all its attached trees. Then

$$\Psi_X = K_0(Z_{C_5}(b) + 2i), \quad b_v \geq t.$$

Coefficientwise monotonicity gives $Z_{C_5}(b) \geq Z_{C_5}(t)$, hence $0 < \Theta_X(t) \leq \Theta_{C_5}(t)$ and

$$D(X) \geq D(C_5) = 4 - 2\sqrt{5}. \quad (26)$$

For the last equality use the eigenvalues $2 \cos(2\pi j/5)$, which give $s^+(C_5) = 7 - \sqrt{5}$. Combining (25)–(26) gives $D(G) > 8 - 4\sqrt{5}$; equation (2) proves the result. $\square$

This is the actual-lobe continuant proof of [6]; it includes the single-bridge case treated separately in *Positive Square Energy for Two Pentagons Joined by a Bridge with Arbitrary Trees* [7]. Comparing with bare lobes would be invalid: attachments can increase their pointwise phase.

5 Exhaustive classification

Theorem 5.1. *Every connected bicyclic cactus G on n vertices satisfies*

$$\boxed{s^+(G) \geq n}.$$

The cycles may share a cut vertex or be disjoint with an arbitrary connector, and arbitrary trees may occur throughout the block incidence structure.

Proof. Let p, q be the two cyclic block lengths. If one is even, or if both are odd and at least five but not both five, (13) and (12) apply. If both are 3 (mod 4), the common-phase argument (14) applies. A remaining pair containing a triangle is exactly $\{3, 4k+1\}$ and is settled by Theorem 3.1. The sole remaining pair $\{5, 5\}$ is settled by Theorem 4.1 when the cycles share a vertex and by Theorem 4.2 when they are disjoint. These exhaust the two possible incidences of cyclic blocks in a cactus and all unordered length pairs. $\square$

The mixed and two-pentagon residual estimates are strict. The weak form in Theorem 5.1 is the uniform conclusion needed for the AKMPZ bound. Nothing here claims the same theorem for non-cactus bicyclic graphs.

A Certificate paths and reproduction

All paths below are relative to the repository root. The bouquet coefficient certificate is checked exactly with

```
python positive-square-energy/experiments/c5_bouquet_matching_certificate.py
```

The expected summary is

```
PASS two-C5 bouquet matching coefficient certificate
terms=1290 min_coefficient=1 max_coefficient=22
y_constant=0 all_coefficients_nonnegative=True
sha256=4c436cac772395d2a8edfdd81408ffe426759d3e94d66df2e4ab0235a3343110
```

The connector factorization has an independent, nonessential audit at

```
python positive-square-energy/experiments/connector_factor_audit.py
```

The source manuscripts used to assemble this proof are

```
sharp-cactus-dnn/paper.tex
triangle-one-mod-four-cactus/paper.tex
two-c5-bouquet-trees/paper.tex
two-c5-all-connectors/paper.tex
two-c5-bridge-trees/paper.tex
```

References

- [1] S. Akbari, H. Kumar, B. Mohar, S. Pragma, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264, 2025.
- [2] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.
- [3] Companion manuscript, *The Optimized Liu–Tang–Zhang Constant for Cactus Graphs*, 2026; repository path `sharp-cactus-dnn/paper.tex`.
- [4] Companion manuscript, *The Triangle- C_{4k+1} Residual Family for Bicyclic Cacti*, 2026; repository path `triangle-one-mod-four-cactus/paper.tex`.

- [5] Companion manuscript, *Positive Square Energy for a Two- C_5 Bouquet with Arbitrary Trees*, 2026; repository path `two-c5-bouquet-trees/paper.tex`.
- [6] Companion manuscript, *Positive Square Energy for Two Pentagons Joined by an Arbitrary Path*, 2026; repository path `two-c5-all-connectors/paper.tex`.
- [7] Companion manuscript, *Positive Square Energy for Two Pentagons Joined by a Bridge with Arbitrary Trees*, 2026; repository path `two-c5-bridge-trees/paper.tex`.